

Statistical Assurance Regions for Predictive Models Under Joint Missingness and Distribution Shift

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Abstract

A frozen predictive pipeline can satisfy a risk requirement in some deployment conditions and fail in others. SciGuard represents this evaluation problem through a statistical assurance region: an estimated inner subset of a prespecified environment family whose predictive risks are bounded by a tolerance. We develop an auditable workflow using established simultaneous inference, with finite-sample exact-binomial and bounded-loss alternatives alongside a dependence-aware Gaussian multiplier band. We state the sampling construction that permits shared stress transformations, distinguish uncertainty in the assurance sample from uncertainty in simulation reference risks, and give an independent-pilot design procedure. A continuous extension is available only with an externally justified global smoothness bound. Experiments include nested predictive-pipeline refits, missingness-mechanism comparisons, a breast-cancer data stress test, and computational and adverse-regime ablations. The rerun exposes limits of Gaussian approximation: multiplier simultaneous coverage is 0.932 in the primary design and 0.775 in a rare-event design at a nominal 0.95 level. Exact-binomial alternatives trade efficiency for finite-sample control. A corrected MIMIC-IV temporal protocol is implemented but has not been executed on credentialed data. The contribution is a scoped methodological and reproducibility framework; broad deployment utility and clinical safety are not established by the present evidence.

Keywords: deployment assurance, distribution shift, missing data, simultaneous inference, risk control

1 Introduction

An acceptable test-set average does not specify the conditions under which a trained model can meet a performance requirement. A hospital, for example, may encounter changes in both its patient population and the availability of recorded measurements. A useful validation report would identify which explicitly modeled combinations remain supported by the data, while distinguishing an observed low error rate from a statistically bounded risk.

We study this reporting problem for a frozen predictive pipeline. Given environments $e_1, \dots, e_K$, predictive risks $R_1, \dots, R_K$, and a tolerance τ , the target is $\mathcal{A}_\tau = \{e_k : R_k \leq \tau\}$. A simultaneous upper band $(U_1, \dots, U_K)$ produces the certified inner estimate $\hat{\mathcal{A}}_\tau = \{e_k : U_k \leq \tau\}$. For example, upper bounds $(.07, .09, .13)$ at tolerance $.10$ certify the first two environments. The third remains uncertified even if its point estimate is below $.10$; it has

not thereby been proven unsafe. This distinction is the basis of the statistical assurance region (SAR).

The inferential devices underlying this construction are established. Exact binomial limits, bounded-sum inequalities, multiple testing, and Gaussian multiplier approximation already provide ways to control joint error (Clopper and Pearson, 1934; Hoeffding, 1963; Holm, 1979; Chernozhukov et al., 2013). Learn then Test (LTT) can directly index its hypotheses by deployment environments (Angelopoulos et al., 2025). Merely naming the resulting set does not create a new inferential theorem. We therefore position SciGuard as an applied methodology and software workflow that connects an explicit environment design, sampling assumptions, uncertainty accounting, and an interpretable set-valued report.

The revision makes four substantive contributions to that workflow. First, it separates finite-sample risk control from asymptotic dependence-aware inference and states when each is appropriate. Second, it gives the row-level construction needed to justify reuse of observations under multiple transformations, including corrections to a missingness generator that previously did not implement the claimed MAR mechanism. Third, it puts nested fitted-model experiments and reference-risk uncertainty at the center of the evaluation. Fourth, it provides one executable notebook and shared software implementation from which all current tables and figures are generated. An additional local MIMIC-IV protocol addresses temporal alignment, patient separation, class imbalance, and missingness reporting, but its empirical execution remains outstanding.

The results do not support universal use of the multiplier method. It can enlarge the certified region when losses across environments are strongly dependent, but its simultaneous coverage deteriorates for sparse errors in the examined finite samples. This finding motivates retaining simple exact or bounded-loss procedures as substantive alternatives rather than weak baselines.

2 Relationship to Existing Work

Risk control and simultaneous inference. Risk-controlling prediction sets and LTT provide model-agnostic calibration and testing frameworks (Bates et al., 2021; Angelopoulos et al., 2025). In our finite-family setting, the hypotheses $H_{0k} : R_k \geq \tau$ are an immediate LTT application. SciGuard’s environment-indexed output is an operational interpretation of this framework, not a disjoint class of guarantees. Simultaneous confidence bands also have a substantial history beyond high-dimensional Gaussian maxima; for example, Degras (2011) studies bands for functional regression. Here the coordinates are prespecified environment-specific predictive risks. Their construction and sampling design, rather than a new general band theorem, are the focus.

Prediction and adaptation under shift. ATC predicts target accuracy from source-calibrated confidence scores and unlabeled target observations (Garg et al., 2022). Its inputs and objective differ from labeled environment certification. We therefore evaluate it using risk-prediction error, not by ranking its “certification” error against a formal familywise procedure. Weighted conformal prediction addresses covariate shift using likelihood-ratio information under its stated conditions (Tibshirani et al., 2019); it does not by itself give our simultaneous region over an arbitrary family of missingness interventions. Missingness-shift

adaptation changes a predictor for target conditions (Zhou et al., 2023). An adapted pipeline could subsequently be frozen and assessed with the same assurance workflow.

Generalization bounds and robust learning. PAC-Bayes analysis controls the generalization of distributions over predictors using complexity terms (Alquier, 2024). Distributionally robust optimization changes the training objective to improve performance over an uncertainty set (Duchi and Namkoong, 2021). Our main route instead conditions on a fitted pipeline and uses independent labeled assurance observations. It makes no claim to dominate these approaches and does not transport labels to unobserved deployment populations without additional assumptions.

Safe operating regions. Dixit et al. (2025) couple conformal perception calibration to a safe planner and evaluate navigation safety. Their guarantee concerns the resulting system, whereas our risk threshold alone says nothing about downstream control or clinical harm. Torens et al. (2025) monitor an operational design domain for an ML component; here uncertainty in average predictive loss determines an inner set within an explicitly supplied family. Hammar et al. (2026) learn a network’s safe operating region through causal observation and intervention. SciGuard’s present empirical analysis is offline and does not implement such active region expansion. These are meaningful precedents for region-based assurance, not evidence that the term SAR denotes a previously unstudied class of problems.

3 Target and Sampling Design

Let D_0 include all training and design information, including preprocessing estimates, learner randomization, a pilot sample if used, and the selected finite environment family $\mathcal{E}(D_0) = \{e_1, \dots, e_K\}$. We condition on D_0 unless otherwise stated. The fitted pipeline g includes imputation, transformations, prediction, and any fixed decision threshold. For bounded loss $\ell \in [0, 1]$, define

$$R_g(e) = \mathbb{E}_{P_e}[\ell(Y, g(X_{\text{obs}}, M))], \quad \mathcal{A}_\tau(g) = \{e \in \mathcal{E}(D_0) : R_g(e) \leq \tau\}. \quad (1)$$

The law P_e must be specified. Synthetic perturbation experiments establish results only for their artificial environment laws, not automatically for a real hospital or calendar period.

3.1 Paired stress experiments

Lemma 1 (Shared transformations) *Conditional on D_0 , let (Z_i, V_i) be i.i.d. across i , independent of D_0 , and let $h_k(\cdot; D_0)$ be fixed measurable maps into $[0, 1]$. Then $\mathbf{L}_i = (h_1(Z_i, V_i; D_0), \dots, h_K(Z_i, V_i; D_0))$ are i.i.d. bounded random vectors. No independence across coordinates is required.*

Proof For any measurable vector sets $B_1, \dots, B_n$, the events $\{\mathbf{L}_i \in B_i\}$ are measurable with respect to their respective independent (Z_i, V_i) rows, conditional on D_0 . Their joint conditional probability factors. The same row law and fixed maps give the same vector distribution for each i . ■

Shared masks, noise, or outcome uniforms within a row are therefore compatible with simultaneous inference. Perfectly correlated coordinates are permitted. In contrast, normalizing missingness probabilities by the assurance-batch mean makes the transformation depend on other rows and is not justified by Lemma 1. Our corrected generators use fixed population-calibrated intercepts in the simulation. Repeated patients, temporal dependence, or cluster sampling also require an appropriate analysis rather than this i.i.d. argument. Conditioning on training data does not remove dependence among assurance rows.

3.2 Natural groups and independent samples

For a master sample of independent subjects, let G_{id} indicate a prespecified domain or subgroup. With $n_d = \sum_i G_{id}$ and $\hat{p}_d = n_d/n$, estimate

$$\hat{R}_{d,m} = \frac{\sum_i G_{id} L_i^{(m)}}{n_d}, \quad \hat{\psi}_{i,d,m} = \frac{G_{id}}{\hat{p}_d} (L_i^{(m)} - \hat{R}_{d,m}). \quad (2)$$

Common row multipliers applied to these influence estimates preserve overlapping-group and within-subject stress dependence. Fixed-dimensional asymptotic ratio inference additionally requires positive group probabilities and nondegenerate influence variances. Small or empty groups remain in the report and cannot certify under the prespecified minimum-size rule. For exact binomial inference, condition on a group's membership indicators: under i.i.d. master rows and a fixed group definition, its selected binary losses have the group-conditional Bernoulli law. A union bound can then cover all groups and stress conditions, despite overlap.

When environments instead have genuinely independent, unequal samples, SciGuard accepts separate loss vectors. It neither truncates all samples to their shortest length nor pairs unrelated patients by row position. Validity remains a statement about the represented environment populations, not unobserved sites.

4 Risk Bounds and Set Containment

4.1 Finite-sample routes

Let $\hat{R}_k$ be the mean of n_k independent bounded losses with expectation R_k . For binary losses, write $S_k = n_k \hat{R}_k$. The one-sided exact binomial bound is

$$U_k^{\text{CP}} = \begin{cases} Q_{1-\alpha/K} \{\text{Beta}(S_k + 1, n_k - S_k)\}, & S_k < n_k, \\ 1, & S_k = n_k, \end{cases} \quad (3)$$

where Q denotes a quantile (Clopper and Pearson, 1934). For general losses in $[0, 1]$, use

$$U_k^{\text{H}} = \min \left\{ 1, \hat{R}_k + \sqrt{\frac{\log(K/\alpha)}{2n_k}} \right\}. \quad (4)$$

Proposition 2 (Finite-sample simultaneous assurance) *Conditional on the frozen design, either (3), for Bernoulli samples, or (4), for independent bounded samples, satisfies*

$$\mathbb{P}\{R_k \leq U_k \text{ for every } k \mid D_0\} \geq 1 - \alpha. \quad (5)$$

Consequently, on this same event, $\hat{\mathcal{A}}_\tau \subseteq \mathcal{A}_\tau$ for every $\tau \in [0, 1]$ simultaneously. Dependence between environment samples is allowed.

Proof An exact one-sided binomial limit fails with probability at most α/K in each coordinate. For (4), the bounded-sum inequality of Hoeffding (1963) gives the same marginal failure bound. The union bound yields (5) without requiring independence between coordinates. On that event, $U_k \leq \tau$ implies $R_k \leq U_k \leq \tau$ for every coordinate and every threshold. ■

These are standard constructions, stated here to make the software guarantee explicit. For example, $S_k = 0$ gives $U_k^{\text{CP}} = 1 - (\alpha/K)^{1/n_k}$, a positive uncertainty bound despite no observed errors. Exact Holm uses $p_k = \mathbb{P}\{\text{Bin}(n_k, \tau) \leq S_k\}$ and the step-down rule of Holm (1979). These are valid lower-tail tests of $H_{0k} : R_k \geq \tau$ and yield finite-sample familywise containment at that fixed tolerance. The method is an environment-indexed LTT instantiation; it need not supply a full upper band.

4.2 Dependence-aware multiplier route

For paired rows, define $\hat{\sigma}_k^2 = n^{-1} \sum_i (L_{ik} - \hat{R}_k)^2$ and

$$T^* = \max_k \left\{ -\frac{1}{\sqrt{n}\hat{\sigma}_k} \sum_{i=1}^n \xi_i (L_{ik} - \hat{R}_k) \right\}, \quad \xi_i \stackrel{\text{iid}}{\sim} N(0, 1). \quad (6)$$

Let $c_{1-\alpha}^*$ be its conditional quantile. The upper band is

$$U_k^{\text{M}} = \min\{1, \hat{R}_k + c_{1-\alpha}^* \hat{\sigma}_k / \sqrt{n}\}. \quad (7)$$

Software enlarges negative simulated critical values to zero and assigns bound 1 to empirically constant columns. This avoids invalid studentization; it is not a general repair for rare-event approximation error.

Proposition 3 (Fixed-family asymptotic justification) *Under Lemma 1, for fixed K and positive population coordinate variances, the ideal conditional quantile in (6) yields $\liminf_{n \rightarrow \infty} \mathbb{P}(R_k \leq U_k^{\text{M}} \text{ for all } k \mid D_0) \geq 1 - \alpha$. For an empirical bootstrap quantile, the statement also requires the number of draws to increase sufficiently for quantile estimation error to vanish.*

Proof The multivariate central limit theorem applies to the independent bounded loss vectors. The sample covariance converges entrywise to the population covariance and each sample standard deviation converges to a positive constant. Conditional on the data, the vector in (6) before taking its maximum is exactly Gaussian with the empirical standardized covariance. Its distribution therefore converges to the same Gaussian limit as the standardized mean errors. The maximum of finitely many Gaussian coordinates with nonzero marginal variance has a continuous distribution, even if the covariance is singular. Quantile consistency and the continuous mapping argument give asymptotic coverage. The set-containment implication then follows as in Proposition 2. ■

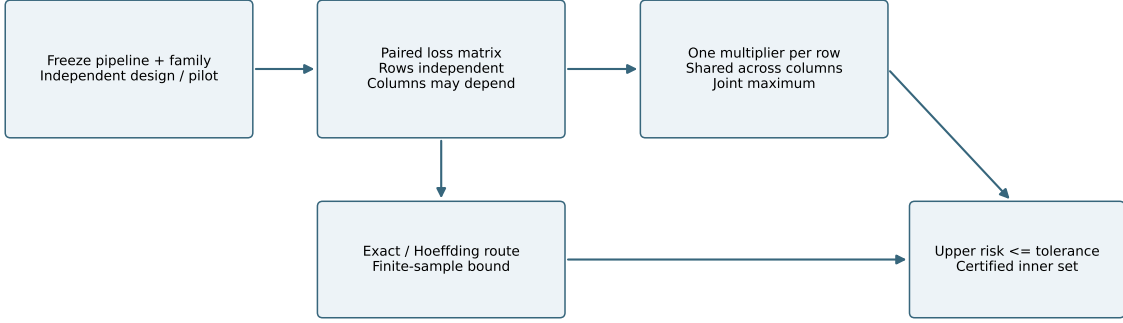


Figure 1: The workflow separates the sampling design from the choice of inferential bound. The multiplier route shares one random weight across all losses in a row. Exact and bounded-loss alternatives reach the same set-valued output with different guarantees.

For increasing K , the high-dimensional results of Chernozhukov et al. (2013) provide sufficient regimes for bounded coordinates with uniformly positive variances; a familiar conservative regime is $\log^7(Kn)/n \rightarrow 0$ under the associated regularity constants. Lemma 1 establishes the relevant independent-vector construction; bounded losses and a uniform variance lower bound supply its basic moment conditions. This rate is not a practical sample-size prescription. Later refinements improve Gaussian and bootstrap approximation rates (Chernozhukov et al., 2022); we do not assert their sharpest rates or optimality for the present grouped and stressed-data implementation. The experiments assess finite-sample behavior directly and show that the assumptions can be practically uninformative for rare errors.

4.3 Conditioning, tuning, and repeated use

If a finite-sample statement holds conditional on every admissible training/design realization, averaging over D_0 preserves its error bound. Nested refits illustrate variation across fitted pipelines; they do not transform an asymptotic method into a finite-sample theorem. Proposition 2 also clarifies a useful distinction: a fixed risk band supports all tolerances simultaneously, including a threshold inspected afterward. That mathematical fact does not justify retroactively defining an operationally meaningful risk requirement. Changing models, losses, families, or choosing the most favorable unadjusted method after assurance is different and requires protection over the enlarged selection problem.

Repeated 95% checks are not a time-uniform 95% statement. A simple conservative extension uses fresh assurance data at round t , conditions on the past, and applies a finite-sample procedure at $\alpha_t = \alpha/[t(t+1)]$. Since $\sum_{t \geq 1} \alpha_t = \alpha$, a union bound controls any erroneous certification over all rounds. This standard spending argument does not supply efficient adaptive or optimal sequential region learning.

5 Environment Design and Continuous Regions

The environment family is a scientific input. Begin with a specified loss and decision threshold, define interpretable axes and externally motivated ranges, and distinguish changes to the population from changes to measurement or observation. A transformation of one labeled sample is not evidence that an intended real deployment law is represented. Omitted axes remain outside the scope of the report.

Our independent-pilot procedure starts with a candidate grid, retains points near a pilot-estimated boundary together with coordinate extremes, and freezes that selected finite family before examining an independent assurance sample. Conditional on the pilot, the exact binomial construction applies without selective-inference machinery. Alternatively, one may first construct a valid band on an entire fixed superset and report a subset of it. Reusing the same outcomes to invent new environments and applying an unadjusted band only afterward is not justified by either argument.

For planning, (4) has additive width at most ϵ when $n \geq \log(K/\alpha)/(2\epsilon^2)$. This bound is conservative and controls band width, not the probability of certifying a cell with a particular true risk. Pilot simulations can estimate recall for plausible risk margins, but they cannot establish external validity.

Suppose a compact continuous family has a scientifically justified global bound $|R(e) - R(e')| \leq L_R d(e, e')$. Define

$$U^{\text{cont}}(e) = \min \left\{ 1, \min_{k \leq K} [U_k + L_R d(e, e_k)] \right\}. \quad (8)$$

On the grid-coverage event, $R(e) \leq R(e_k) + L_R d(e, e_k) \leq U_k + L_R d(e, e_k)$ for every k , so (8) covers the risk everywhere. Thresholding it gives a continuous inner region. This is a standard smoothness consequence of simultaneous inference.

The largest observed slope on a finite grid is not a valid global upper bound on L_R . An independent pilot removes selection bias but cannot exclude unobserved rapid variation between design points without structural assumptions. If an externally justified random $\hat{L}_R$ bounds the global constant with failure probability δ , the combined error budget is at most $\alpha + \delta$. Without such a bound, finite-grid certification and explicitly assumption-dependent smoothness sensitivity are the defensible outputs.

6 Experimental Design and Evaluation

All reported results are generated by the revised shared implementation, with seed 20260927. They supersede the historical CSVs retained under `archive/pre-revision/`. Models, designs, and inferential settings are fixed in code; outcomes are not filtered to match earlier published-looking numbers. The full profile and smoke profile write separate directories. Tables and numerical macros are exported from full-run CSVs with a hash-based provenance record.

With known risk vector R , containment is $C = \mathbb{I}\{\hat{\mathcal{A}}_\tau \subseteq \mathcal{A}_\tau\}$, ARR is $|\hat{\mathcal{A}}_\tau \cap \mathcal{A}_\tau|/|\mathcal{A}_\tau|$, and UCP is $|\hat{\mathcal{A}}_\tau \setminus \mathcal{A}_\tau|/|\hat{\mathcal{A}}_\tau|$. We set UCP to zero for an empty certified set and leave ARR undefined when the true safe set is empty. Counts accompany these metrics. A high containment frequency can be vacuous when almost nothing is certified, and it does not

imply that a full risk band covers at its advertised confidence level. Wilson intervals describe Monte Carlo uncertainty in binary coverage/containment indicators; mean recalls and paired differences have replication-level Monte Carlo standard errors (MCSEs).

6.1 Nested fitted-pipeline study

The central predictive-pipeline experiment uses 200 independent refits. Each generates 1,200 training rows, 400 source-calibration rows for ATC, 1,000 assurance rows, and 20000 separate reference rows. Ten Gaussian covariates have AR(1) covariance with parameter .35. Outcomes follow

$$\mathbb{P}(Y = 1 \mid X) = \text{expit} \left[1.5 \{ .75X_0 - .65X_1 + .45X_2 + .55X_0X_1 - .35X_2^2 + .20X_3 \} \right]. \quad (9)$$

A fixed normalized direction $d \propto (.8, -.6, .4, .2, 0, \dots, 0)$ induces mean shift $s\Sigma^{1/2}d$, using the Cholesky factor. The grid has $m \in \{0, .1, .2, .3\}$ and $s \in \{0, .15, .30, .45\}$, with $\tau = .35$. Feature X_0 remains observed for comparability with the MAR study; severity m is the missingness rate among the other nine features. Imputation uses training means. The two pipelines are standardized logistic regression and histogram gradient boosting (60 boosting iterations, 15 maximum leaves, learning rate .08). The same base rows, label uniforms, and missingness randomness are used across models and environments.

The independent reference mean $\tilde{R}_k$ is not called an oracle truth. Simultaneous two-sided exact binomial intervals $[a_k, b_k]$ allocate total error .01 over both models' reference risks within each refit. Define confirmed-safe cells by $b_k \leq \tau$, confirmed-unsafe cells by $a_k > \tau$, and all others as ambiguous. Then

$$\begin{aligned} C^- &= \mathbb{I}\{\text{all certified cells are confirmed safe}\}, \\ C^+ &= \mathbb{I}\{\text{no certified cell is confirmed unsafe}\}. \end{aligned} \quad (10)$$

satisfy $C^- \leq C \leq C^+$ on that reference-coverage event. The means of these endpoints are reported as a reference bracket, not as confidence limits for a coverage probability. Point-reference containment and ARR remain labeled as such. The reference interval budget is per refit; it is not a joint .99 guarantee over all refits.

Table 1 gives point-reference containment 1.000 for all methods, which must not be read as proof of perfect containment. Logistic-regression cells near the boundary remain ambiguous under the reference intervals; the reference bracket is wider for the multiplier procedure. HGB has no ambiguous reference cells in these runs, making its favorable recall a result for this particular fitted-model regime rather than evidence about difficult near-boundary clinical cases. The relative advantage over exact Holm is also model-dependent: the two methods have nearly equal HGB recall. Figure 2 makes these differences visible without assigning ATC a certification interpretation.

ATC uses maximum class probability and a source-calibrated strict threshold, with tied scores handled jointly. Table 2 reports its risk-prediction MAE, signed bias, and RMSE against the independent reference means. No target labels enter its risk predictions, whereas the certification methods use labeled assurance losses. This is a descriptive neighboring task, not a head-to-head comparison of equal guarantees.

Table 1: Nested refits. LR denotes logistic regression and HGB histogram gradient boosting. The reference bracket averages (10); it captures reference-risk ambiguity, not Monte Carlo uncertainty in a population coverage probability. ARR uses reference means.

Model	Method	Reference containment [95% CI]	ARR	Reference bracket
LR	Bonferroni normal	1.000 [0.981, 1.000]	0.112	[0.990, 1.000]
LR	Multiplier	1.000 [0.981, 1.000]	0.178	[0.980, 1.000]
LR	Bonferroni exact	1.000 [0.981, 1.000]	0.086	[1.000, 1.000]
LR	Hoeffding	1.000 [0.981, 1.000]	0.011	[1.000, 1.000]
LR	Exact Holm	1.000 [0.981, 1.000]	0.102	[0.985, 1.000]
HGB	Bonferroni normal	1.000 [0.981, 1.000]	0.666	[1.000, 1.000]
HGB	Multiplier	1.000 [0.981, 1.000]	0.743	[1.000, 1.000]
HGB	Bonferroni exact	1.000 [0.981, 1.000]	0.624	[1.000, 1.000]
HGB	Hoeffding	1.000 [0.981, 1.000]	0.380	[1.000, 1.000]
HGB	Exact Holm	1.000 [0.981, 1.000]	0.739	[1.000, 1.000]

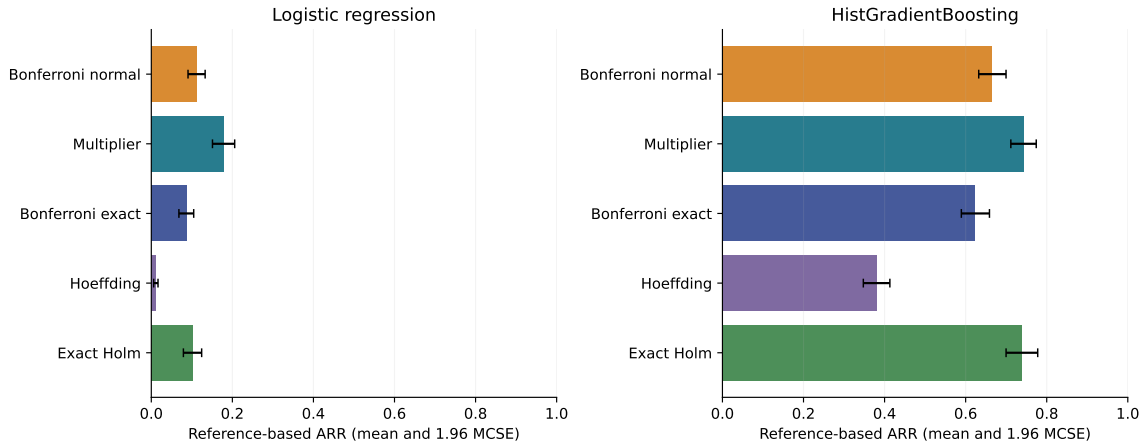


Figure 2: Reference-based recall across nested refits. Bars are means; error bars are 1.96 MCSE. These error bars describe Monte Carlo variation, not uncertainty in the reference risks.

6.2 Missingness-mechanism study

Across 100 logistic-regression refits, we compare MCAR, MAR, MNAR, and structured missingness with the same training and evaluation design. MAR depends only on always-observed X_0 ; MNAR depends on the latent value of the feature being hidden. Logistic intercepts solve a population Gaussian expectation equation for the target marginal missingness, using 64-node Gaussian quadrature. They are fixed before evaluation and are not normalized using assurance rows. Structured failure makes features 1–3 missing together, with every eligible feature retaining marginal severity m . The original cyclic driver scheme could hide its own conditioning variables and was not a valid general MAR implementation.

Table 2: ATC-MC performance prediction, averaged over nested refits.

Model	MAE	Bias	RMSE
Logistic regression	0.027	-0.007	0.029
HistGradientBoosting	0.030	-0.016	0.033

Table 3: Multiplier results for missingness mechanisms. Ambiguous cells are averaged over the 16 environments per refit. All metrics use the corrected mechanism definitions.

Mechanism	Reference containment	Reference bracket	ARR	Ambiguous cells
MCAR	1.000 [0.963, 1.000]	[0.890, 1.000]	0.244	2.12
MAR	1.000 [0.963, 1.000]	[0.960, 1.000]	0.287	0.44
MNAR	1.000 [0.963, 1.000]	[1.000, 1.000]	0.356	0.02
Structured	1.000 [0.963, 1.000]	[0.960, 1.000]	0.271	1.26

Reference intervals allocate error .01 over all four mechanisms within each refit. Table 3 shows that the reference uncertainty can matter even when every point-reference containment indicator equals one. In particular, ambiguous cells reduce the MCAR multiplier lower endpoint. These results concern the chosen parameterizations; they do not imply that MNAR is generally easier or safer than MCAR. Realized missingness fractions and all exact-baseline comparisons are retained in the current-run CSVs.

6.3 Calibration with exactly known risks

To isolate inference, a Gaussian copula generates binary losses with the 25 marginal risks stored in the supplied simulation surface, $n = 1000$, latent correlation .55, and $\tau = .34$. We run 500 replications and 2,000 multiplier draws per replication. This is a calibration experiment, not a fitted-pipeline deployment study. All methods receive the same loss matrix in each replication.

Pointwise normal intervals attain only 0.514 simultaneous coverage. The multiplier band attains 0.932, compared with 0.960 for exact Bonferroni (Table 4). The multiplier’s mean ARR is 0.298, compared with 0.258 for exact Bonferroni. This efficiency difference accompanies different finite-sample validity: the multiplier is not a finite-sample replacement for exact inference. Its primary coverage estimate lies below .95; the Monte Carlo interval describes the precision of that finding and does not verify a universal nominal guarantee.

6.4 Dependence, sample size, and adverse regimes

The sensitivity study uses 300 replications per setting, varying $n \in \{500, 1000, 5000\}$ at latent correlation .55 and correlation $\rho \in \{.10, .55, .90\}$ at $n = 1000$. Table 8 reports paired recall gains, not differences between independently simulated studies. Against normal Bonferroni the gain is negligible at low correlation and larger at high correlation. Comparison with exact Holm is necessary: exploiting dependence is not the only way to reduce multiplicity conservatism.

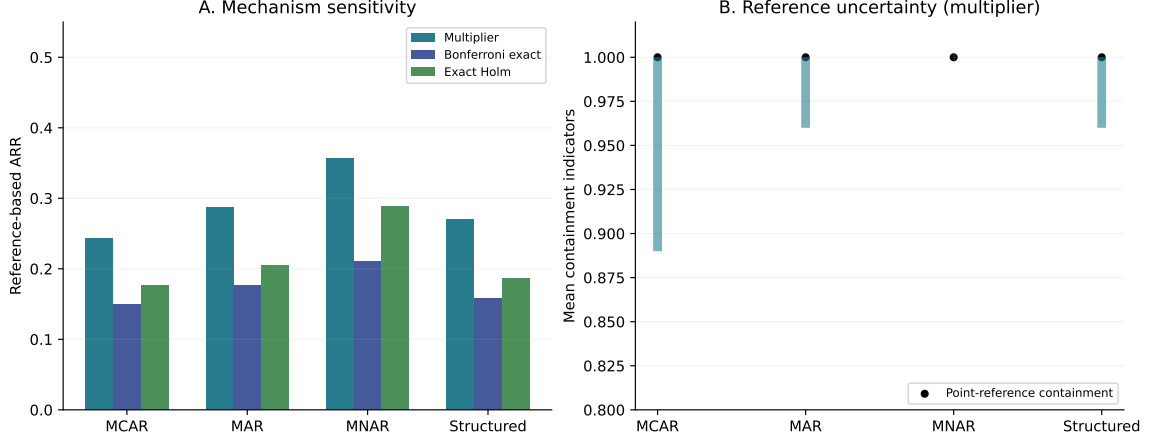


Figure 3: Mechanism sensitivity and the effect of reference-risk uncertainty. Vertical segments in panel B span mean lower/upper containment indicators; they are not confidence intervals.

Table 4: Primary calibration study. Coverage is simultaneous risk-band coverage. Holm supplies fixed-tolerance tests rather than a risk band, so its coverage entry is inapplicable.

Method	Coverage [95% CI]	Containment [95% CI]	ARR	UCP
Pointwise normal	0.514 [0.470, 0.558]	0.958 [0.937, 0.972]	0.522	0.0042
Bonferroni normal	0.938 [0.913, 0.956]	0.998 [0.989, 1.000]	0.282	0.0002
Multiplier	0.932 [0.906, 0.951]	0.998 [0.989, 1.000]	0.298	0.0002
Bonferroni exact	0.960 [0.939, 0.974]	0.998 [0.989, 1.000]	0.258	0.0002
Hoeffding	1.000 [0.992, 1.000]	1.000 [0.992, 1.000]	0.138	0.0000
Exact Holm	—	0.998 [0.989, 1.000]	0.270	0.0002

Three additional 200-replication settings challenge the main result: independent Beta losses with concentration four and known means from the main surface; binary risks equally spaced from .002 to .08 at $n = 500$, $K = 25$, and $\tau = .035$; and a $K = 125$ binary family at $n = 1000$ formed by repeating the main risk values with new copula-specific noise. The last setting enlarges the number of hypotheses, not the dimensionality of continuous deployment geometry. Losses remain bounded; no robustness claim is made for unbounded heavy tails or adversarially selected environments.

The rare-event multiplier coverage is 0.775 (Table 5). Refusing to certify zero-variance columns does not prevent this failure in low-count, nonconstant columns. Exact binomial methods remain applicable, and Hoeffding can be very conservative. These adverse results provide a practical reason to prefer a finite-sample route when such control is required. Selecting whichever unadjusted bound yields the most certificates after seeing assurance losses is not a valid remedy.

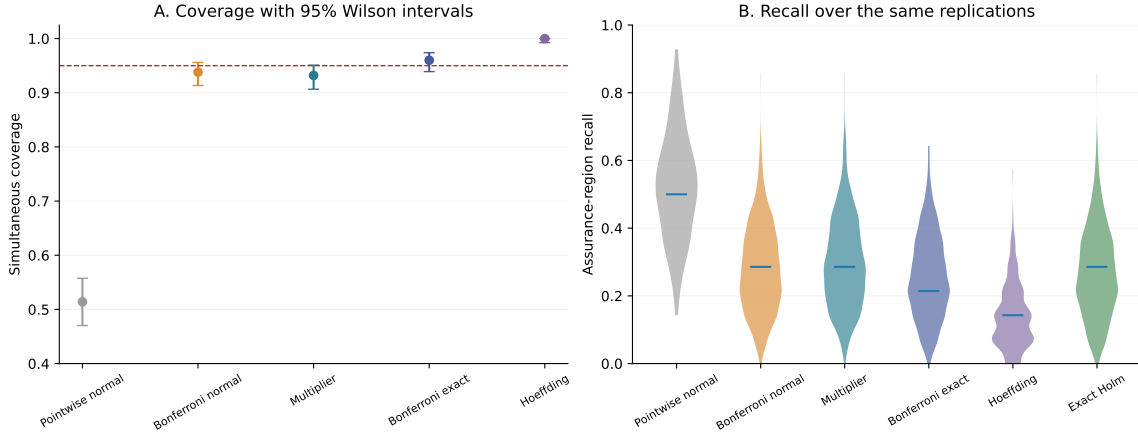


Figure 4: Simultaneous coverage and the recall distribution over primary replications. The dashed line is the nominal coverage target; higher recall alone does not establish validity.

Table 5: Adverse-regime results. Exact binomial bounds are not applicable to continuous losses.

Scenario	Method	Coverage [95% CI]	Containment	ARR
Continuous bounded	Bonferroni normal	0.925 [0.880, 0.954]	1.000	0.469
Continuous bounded	Multiplier	0.925 [0.880, 0.954]	1.000	0.472
Continuous bounded	Hoeffding	1.000 [0.981, 1.000]	1.000	0.008
Rare binary	Bonferroni normal	0.780 [0.718, 0.832]	0.995	0.455
Rare binary	Multiplier	0.775 [0.712, 0.827]	0.990	0.502
Rare binary	Bonferroni exact	0.995 [0.972, 0.999]	1.000	0.379
Rare binary	Hoeffding	1.000 [0.981, 1.000]	1.000	0.000
Larger binary family	Bonferroni normal	0.945 [0.904, 0.969]	0.995	0.208
Larger binary family	Multiplier	0.910 [0.862, 0.942]	0.990	0.232
Larger binary family	Bonferroni exact	0.960 [0.923, 0.980]	0.995	0.188
Larger binary family	Hoeffding	0.995 [0.972, 0.999]	1.000	0.088

6.5 Continuous example and independent pilot

The illustrative surface $R(m, s) = .24 + .12m + .10s + .05ms$ on $[0, .4] \times [0, .6]$ has a valid Euclidean Lipschitz bound $L_R = \sqrt{.15^2 + .12^2}$. We use a 5×5 assurance grid, $\tau = .32$, 200 replications, and an 81×81 mesh to compute numerical recall and containment diagnostics. Table 6 distinguishes grid coverage from mesh containment. The analytical argument for (8), not the dense mesh, supplies coverage over every point when the grid band and the global bound are valid. Separate exports evaluate sensitivity to half and twice the analytic bound; the half-bound is not asserted valid.

The pilot example starts with an 11×11 candidate grid on the same rectangle. Using 300 independent pilot rows, it retains points whose pilot risks are within .015 of .32 and

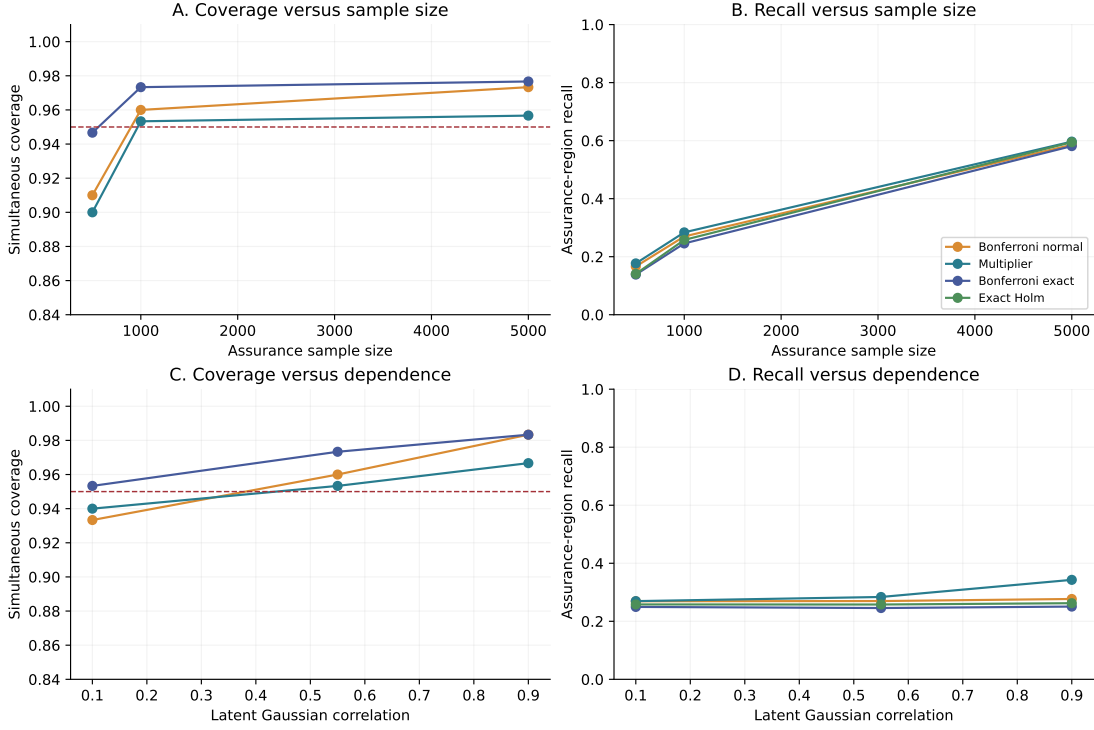


Figure 5: Coverage and recall sensitivity. Latent Gaussian correlation is a generator parameter, not the observed binary-loss correlation. Holm is omitted from risk-band coverage panels.

Table 6: Continuous example. Mesh recall is an equal-grid-point area approximation.

Method	Grid coverage	Mesh containment	Mesh recall	Mesh UCP
Multiplier	0.945	0.975	0.459	0.0003
Bonferroni exact	0.965	0.980	0.404	0.0001
Hoeffding	1.000	1.000	0.185	0.0000

coordinate extrema, then certifies the selected family on new assurance rows with exact Bonferroni. Two hundred replications and a sample-size planning table are exported. This demonstrates a valid split-sample workflow; it does not establish an optimal adaptive grid or certification of all unselected points.

7 Wisconsin Diagnostic Breast Cancer Stress Test

The supplied WDBC data contain 569 observations. A fixed stratified 65/35 split produces 369 training rows and 200 assurance rows. The frozen models are standardized logistic regression, an Extra Trees ensemble with 500 trees, and histogram gradient boosting. Training means impute missing values; Gaussian feature noise is scaled by training standard

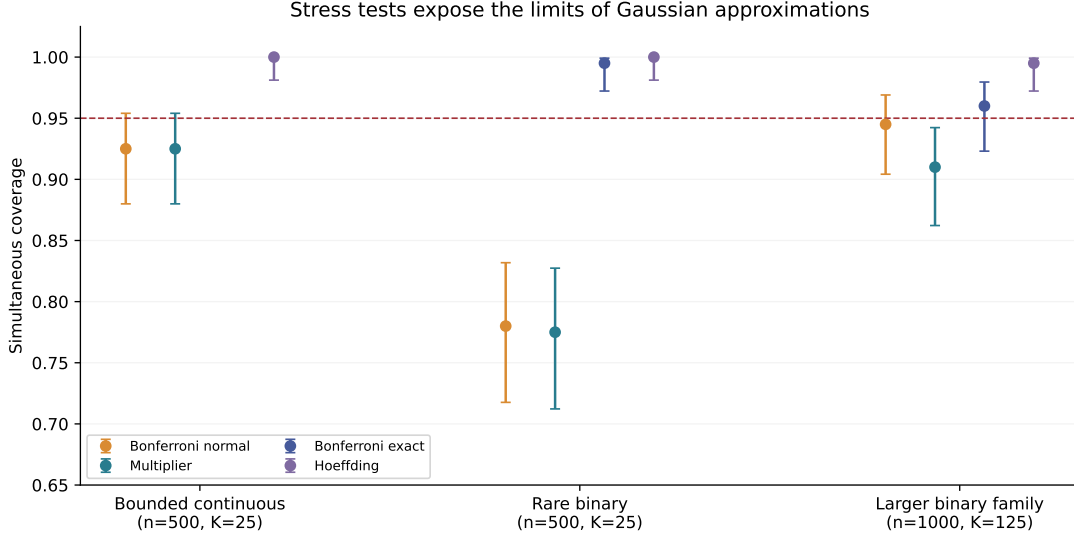


Figure 6: Coverage failures in adverse regimes, with 95% Wilson intervals.

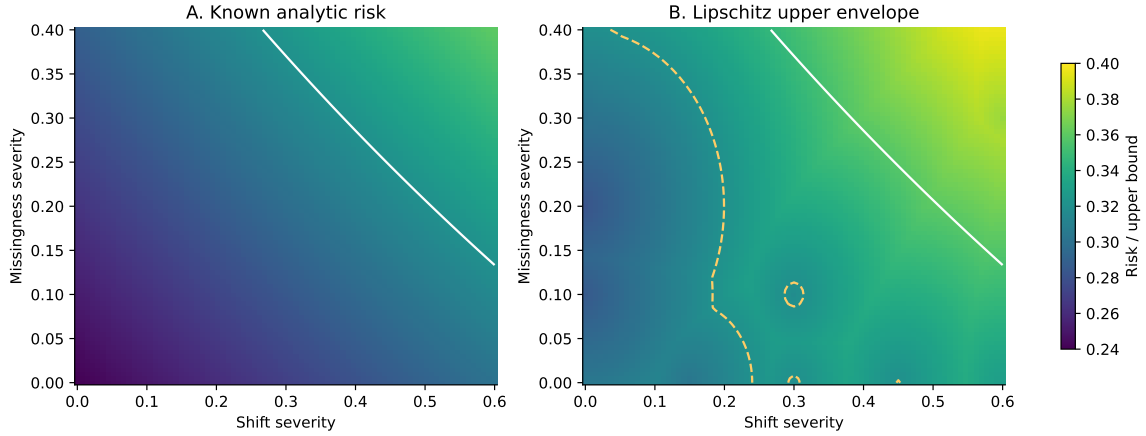


Figure 7: One fixed replication of the continuous example. White denotes the true tolerance boundary; the dashed gold curve denotes the displayed surface’s tolerance contour.

deviations. One noise matrix and one uniform missingness matrix are shared across stress conditions and models. The primary family uses $m \in \{0, .05, .10, .20, .30, .40\}$ and $s \in \{0, .10, .20, .30, .40, .50\}$. The multiplier uses 5,000 draws. Table 7 includes exact procedures so that Gaussian and finite-sample answers are not conflated in this small sample.

Logistic regression has the lowest nominal observed error in this split, but counts depend strongly on the method and tolerance. These are separate model-specific simultaneous statements: choosing a model by comparing these bands does not confer joint model-selection coverage. The tolerances are illustrative, not medical standards. WDBC perturbations are

Table 7: WDBC results for separate model-specific 36-environment families. The nominal observed risk is descriptive. Counts at two tolerances do not establish clinical safety.

Model	Method	Nominal risk	Certified $\tau = .08$	Certified $\tau = .10$
LR	Multiplier	0.010	23/36	30/36
LR	Bonferroni normal	0.010	20/36	29/36
LR	Bonferroni exact	0.010	15/36	23/36
LR	Exact Holm	0.010	15/36	27/36
Extra Trees	Multiplier	0.035	9/36	19/36
Extra Trees	Bonferroni normal	0.035	2/36	17/36
Extra Trees	Bonferroni exact	0.035	0/36	9/36
Extra Trees	Exact Holm	0.035	0/36	9/36
HGB	Multiplier	0.045	2/36	14/36
HGB	Bonferroni normal	0.045	0/36	12/36
HGB	Bonferroni exact	0.045	0/36	2/36
HGB	Exact Holm	0.045	0/36	2/36

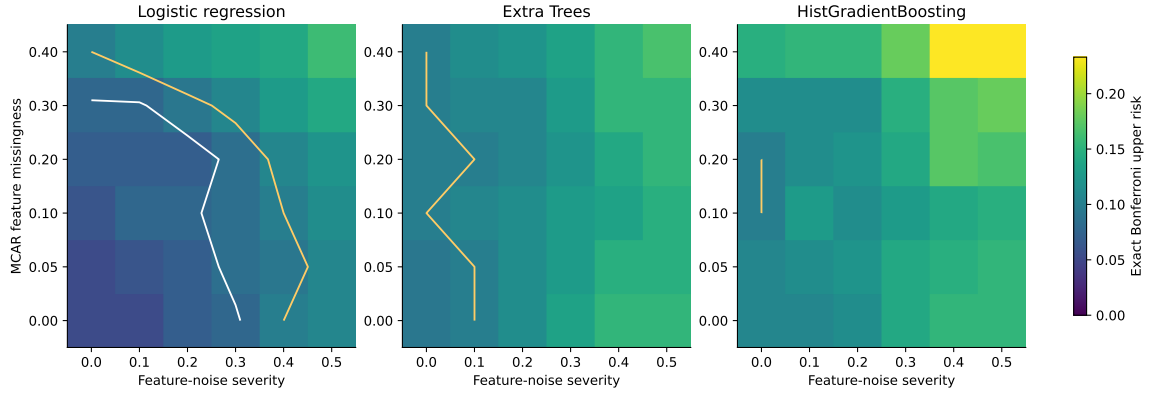


Figure 8: Exact Bonferroni risk surfaces for the three models. White and gold contours, where present, mark tolerances .08 and .10. Absence of a contour can mean no grid crossing.

synthetic and do not represent observed temporal or inter-hospital shifts; its single split cannot estimate the true deployment safe set, ARR, or a coverage probability.

Equally spaced 4×4 , 6×6 , and 10×10 logistic grids span the same rectangle. Their results appear in Table 9. The primary 36-cell grid is nonuniform in missingness and is not identical to the equally spaced 6×6 grid. Certified grid fractions cannot be interpreted as continuous areas without an additional design measure or smoothness argument. A grid is not selected retrospectively for the largest certificate.

8 MIMIC-IV: Implemented Protocol and Remaining Evidence

No MIMIC-IV results are claimed in this revision. The supplied archive contains neither credentialed MIMIC tables nor a completed benchmark output. The earlier primary notebook stopped at its MIMIC section with a missing-file error. We provide a corrected local execution path and validate its mechanics using synthetic table fixtures; that test is not clinical evidence. A genuine naturally shifted evaluation remains necessary to address the central empirical limitation of the present manuscript.

The protocol targets in-hospital mortality for the first eligible adult admission per patient, using admission-table variables only. It excludes identifiers, outcome flags, discharge or death times, and length of stay from predictors. Age top-coding is retained explicitly. The field-availability audit for the intended admission-time decision must still be completed on the actual data; a demographic/admission model is a limited baseline, not a comprehensive clinical prediction system.

MIMIC-IV shifts dates separately for each patient. Its documentation instructs users to align stays to the anchor year rather than treat a patient’s `anchor_year_group` as the period of every admission (Johnson et al., 2024, 2023). The loader adds the shifted admission-year offset to both group endpoints and, by default, pads the approximate interval by one year for year-boundary ambiguity. This padding is an analysis choice, not an official exact-date guarantee. Intervals are restricted to the declared dataset coverage. Source intervals must end by the prespecified cutoff and deployment intervals begin afterward; overlapping intervals are excluded. Deployment groups are approximate admission intervals, which may overlap one another in calendar time, rather than exact years or distinct hospitals.

The example configuration describes v3.1, with source ending by 2016 and subsequent deployment intervals. Users must supply the actual version, scientifically justified tolerance, and rationale before running. The former illustrative .35 mortality-error tolerance is not carried forward as a meaningful clinical requirement. Unweighted logistic regression and histogram gradient boosting are fitted on source patients, with a source holdout kept separate. An always-survive classifier is included. All models, groups, and missingness conditions form one family for the clinical certification comparison, avoiding unadjusted selection across models.

The protocol reports mortality prevalence, sensitivity, specificity, balanced accuracy, AUROC, and AUPRC, because low misclassification risk alone can be misleading for rare outcomes. These diagnostics do not inherit the 0–1-risk guarantee. Native feature missingness is reported separately from synthetic MCAR additions and total missingness. The grouped influence construction (2), exact group-conditional binomial bounds, Hoeffding, and Holm accommodate unequal group sizes without fabricating row alignment. Small groups are retained as uncertified. One admission per patient addresses repeated-patient rows but does not establish independence across time or hospital processes.

Only aggregate tables, the frozen configuration, and source-file hashes are exported by the local protocol. No patient-level cohort or fitted model is included in this revision. Completing the external evaluation requires inspecting actual cohort flow, temporal support, native missingness, label and feature timing, prevalence, trivial-baseline performance, and effective sample sizes before any benchmark result is promoted into the manuscript.

9 Practical Interpretation and Limitations

Choosing a method. Use an exact binary-loss procedure when finite-sample control is central and the independent sampling model is defensible; use a bounded-loss inequality for other bounded losses when conservatism is acceptable. A multiplier band can be useful when the family is strongly dependent and validation supports the approximation, but the rare-event experiment shows that sample size alone is not a sufficient diagnostic. Compare paired recall gains with exact Holm as well as normal Bonferroni, and inspect validity at the same time. The method should be fixed in advance or multiplicity should cover any subsequent method selection.

What is established. We give a coherent assurance target, transparent sampling and inference contracts, a standard finite-sample route, a justified fixed-family multiplier construction, corrected mechanism simulations, explicit reference uncertainty, and reproducible comparisons. The framework accepts any frozen pipeline yielding bounded losses, but the experiments cover a small number of tabular model classes. Interface generality is not empirical evidence for arbitrary neural architectures or deployment systems.

What remains open. The inferential components do not constitute a new optimality theorem. No minimax recovery rate, general safe exploration guarantee, or efficient time-uniform adaptive grid is proved. The continuous extension depends on external smoothness. Unknown shifts and causal transportability remain outside the certificate. Unbounded heavy-tailed losses and adversarial adaptation are not studied. Reference-risk uncertainty persists near the tolerance even with 20,000 rows. The primary simulation deliberately calibrates inference, and WDBC remains small and semisynthetic. The unexecuted MIMIC protocol cannot fill the missing natural-shift evidence. These limitations prevent a claim of general deployment readiness.

10 Conclusion

SciGuard connects a prespecified deployment family to an explicit, auditable set-valued risk report. Established finite-sample and bootstrap tools provide different routes to that report, with different assumptions and efficiency. The revised experiments show both the benefit of accounting for dependence and consequential Gaussian-approximation failures. The appropriate conclusion is a scoped applied framework with reproducible evidence and an implementable external-validation protocol. Stronger claims require completed natural-shift evaluation and, if theoretical novelty is the objective, results beyond the standard simultaneous-inference and smoothness arguments used here.

Data and Code Availability

The accompanying archive contains the canonical MIMIC-IV analysis notebook, the shared Python package, tests, current-run CSVs, generated figures, manuscript sources, and a point-by-point revision response. Historical files are clearly separated. The WDBC CSV and simulation surface are supplied; MIMIC-IV remains credentialed and is not bundled. A public repository and archival identifier are not supplied in this revision. They should be established, and author/funding statements verified, before journal submission.

Appendix A. Additional Tables

Table 8: Paired mean ARR gain of multiplier over each comparator. MCSE uses paired replication differences. These efficiency comparisons must be read with coverage results.

n	ρ	Comparator	Paired ARR gain	MCSE
500	0.55	Bonferroni normal	0.010	0.0017
1000	0.10	Bonferroni normal	0.000	0.0010
1000	0.55	Bonferroni normal	0.014	0.0019
1000	0.90	Bonferroni normal	0.066	0.0039
5000	0.55	Bonferroni normal	0.011	0.0016
500	0.55	Bonferroni exact	0.039	0.0030
1000	0.10	Bonferroni exact	0.020	0.0020
1000	0.55	Bonferroni exact	0.038	0.0031
1000	0.90	Bonferroni exact	0.092	0.0042
5000	0.55	Bonferroni exact	0.015	0.0018
500	0.55	Exact Holm	0.036	0.0029
1000	0.10	Exact Holm	0.011	0.0016
1000	0.55	Exact Holm	0.026	0.0025
1000	0.90	Exact Holm	0.080	0.0041
5000	0.55	Exact Holm	0.002	0.0011

Table 9: Equally spaced WDBC logistic-regression grid ablation.

Grid	Method	K	Certified $\tau = .08$	Fraction
4x4	Multiplier	16	10	0.625
4x4	Bonferroni exact	16	7	0.438
6x6	Multiplier	36	23	0.639
6x6	Bonferroni exact	36	11	0.306
10x10	Multiplier	100	69	0.690
10x10	Bonferroni exact	100	23	0.230

Appendix B. Computational Cost and Bootstrap Variability

The direct multiplier sampler requires $O(BnK)$ operations. Sampling from its conditional Gaussian covariance instead costs $O(nK^2 + K^3 + BK^2)$ and stores a $K \times K$ covariance; it can be advantageous for moderate K . The code chooses that route when $K \leq \min(n, 2000)$ and retains a chunked row-multiplier route otherwise. Conditional distributional equivalence is tested with identical-column and independent-column constructions. Numerical agreement between samplers does not verify frequentist coverage.

The timing study holds the loss matrix fixed and repeats each bootstrap size under 25 independent seeds. Table 10 separates the number of observations, environments, and bootstrap draws. Timings depend on machine and numerical libraries and exclude loss generation/model fitting. The critical-value MCSE is the standard error of its mean across

seeds, not a guaranteed quantile error bound. Per-seed critical values and certified counts are also exported so bootstrap variability is inspectable directly.

Table 10: Conditional-covariance multiplier timing and variability.

n	K	B	Critical mean	Critical MCSE	Mean seconds
500	25	200	2.827	0.0223	0.0005
500	25	1000	2.804	0.0086	0.0008
500	25	5000	2.801	0.0046	0.0024
1000	25	200	2.815	0.0202	0.0006
1000	25	1000	2.781	0.0104	0.0017
1000	25	5000	2.788	0.0041	0.0027
1000	125	200	3.257	0.0227	0.0035
1000	125	1000	3.213	0.0097	0.0050
1000	125	5000	3.210	0.0035	0.0133

Appendix C. Planning and Verification

Table 11: Conservative assurance-sample planning for a Hoeffding additive width at most the stated margin, with $\alpha = .05$. This is not a guarantee of a particular recall.

K	Margin	Sufficient n
25	0.02	7769
25	0.05	1243
25	0.10	311
100	0.02	9502
100	0.05	1521
100	0.10	381
500	0.02	11513
500	0.05	1843
500	0.10	461

Tests cover finite-sample binomial error by enumerating possible counts, the zero-error bound, empirical degeneracy, duplicated columns, covariance/row multiplier equivalence, unequal samples, overlapping and empty groups, reference boundaries, MAR row locality, ATC ties, and admission-aligned MIMIC cohort construction. The synthetic MIMIC fixture exercises the complete local pipeline and aggregate-output schema. It is discarded after testing and is never treated as a benchmark dataset. The canonical notebook imports these same functions, records configuration and package versions, and generates its manuscript inputs from the completed full run rather than hard-coded target values.

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